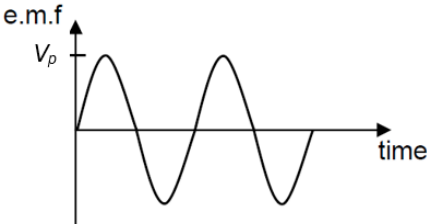
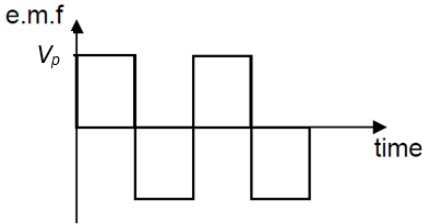


27	A voltage supply connected to a signal generator can either produce an alternating voltage with a sinusoidal waveform or a square waveform with the same peak value of e.m.f. V_p as illustrated below. The voltage supply is connected across a load resistor.   What is the ratio $\frac{\text{mean power dissipated in the load resistor by the square waveform}}{\text{mean power dissipated in the load resistor by the sinusoidal waveform}}$?
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A 0.25

B 0.5

C 2

D 4

L2 Answer: C

$$\langle P \rangle = \frac{v_{rms}^2}{R}$$

For sinusoidal waveform,

$$\langle P \rangle = \frac{\left(\frac{v_p}{\sqrt{2}}\right)^2}{R} = \frac{v_p^2}{2R}$$

For square waveform, using graphical method,

$$v_{rms} = v_p$$

$$\langle P \rangle = \frac{v_p^2}{R}$$

Ratio = 2